Program Operations Analytics

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Role: Independent Analytics Project (Consulting-style)
Tools: Python (pandas), MySQL (SQL), Tableau (dashboard)

Privacy note: This notebook reflects my original code/workflow of the project. The dataset used here is synthetic and generated for privacy (N = 1,523 students across 4 schools).

Executive Summary

The code and workflow in this work sample are drawn from my prior operations analytics work. To protect privacy, the dataset used here is synthetic and was generated to mirror the structure of the original data (N = 1,523 students across 4 schools). The goal is to show how an analyst can move from raw operational tables to **decision-ready reporting**:

- **KPI definition & monitoring:** Weekly engagement and service delivery metrics were defined and aggregated into a dashboard-ready table (scheduled sessions, attended sessions, attendance rate).
- **Cohort retention analysis:** Retention was operationalized at **4-week and 8-week** milestones based on enrollment start/end dates, summarized by cohort month and school to highlight early persistence patterns.
- **Early risk flagging:** A transparent, rule-based risk logic was applied using recent attendance windows (last 2 weeks and last 4 weeks) to classify students into **High / Medium / Low** risk with an explainable "risk reason" field.

Key takeaways (from the dashboard):

1. Overall attendance is **generally stable** with modest variation over time (weekly attendance rate approximately **81%-87%**).
2. Scheduled vs attended service volume shows a consistent gap, indicating predictable no-show/absence patterns and opportunities for operational improvements (e.g., reminders, scheduling adjustments).

3. Risk distribution varies by school; School **1** shows the highest concentration of high-risk flags, which can guide targeted outreach and resource allocation.
4. Cohort retention remains relatively strong (4-week retention roughly **83%-94%** across cohorts), with some cohort/school combinations performing lower than peers.

Deliverables produced:

- `kpi_weekly.csv` for KPI trend monitoring
- `cohort_retention.csv` for cohort retention reporting
- `student_risk.csv` for early warning and outreach workflows
- Tableau dashboard (trend + volume + risk + cohort heatmap)

In [2]:
```
from pathlib import Path
import pandas as pd
import numpy as np

DATA_DIR = Path("data")
OUT_DIR = DATA_DIR

students = pd.read_csv(DATA_DIR / "students.csv", parse_dates=["entry_date", "cohort_month"])
enrollments = pd.read_csv(DATA_DIR / "enrollments.csv", parse_dates=["enroll_date", "exit_date"])
att = pd.read_csv(DATA_DIR / "attendance_weekly.csv", parse_dates=["week_start_date"])
```

Exploratory Data Analysis (EDA)

Before building reporting tables, I ran quick EDA to understand:

- scale (students, enrollments, weekly rows)
- distributions (attendance rates, scheduled sessions)
- variation by school / program track
- retention/exits over time (early exits) These checks help confirm KPI definitions and ensure the dashboard reflects realistic operational patterns.

In [6]:
```
# --- Basic shape ---
print("students:", students.shape)
print("enrollments:", enrollments.shape)
print("attendance_weekly:", att.shape)

# --- Time ranges ---
print("entry_date range:", students["entry_date"].min(), "to", students["entry_date"].max())
print("attendance week range:", att["week_start_date"].min(), "to", att["week_start_date"].max())

# --- School mix ---
display(students["school_id"].value_counts().sort_index())
display(students["program_track"].value_counts())
```

```
students: (1523, 6)
enrollments: (1523, 7)
attendance_weekly: (19521, 5)
entry_date range: 2025-08-01 00:00:00 to 2025-12-15 00:00:00
attendance week range: 2025-07-28 00:00:00 to 2026-03-30 00:00:00
school_id
1    521
2    424
3    326
4    252
Name: count, dtype: int64
program_track
Tutoring      771
Counseling    382
Enrichment    370
Name: count, dtype: int64
```

- The dataset contains 1,523 students, 1,523 enrollments, and 19,521 weekly attendance records. Student entry dates span 2025-08-01 to 2025-12-15, with weekly attendance tracked from 2025-07-28 to 2026-03-30.
- Student volume varies by school: School 1 (521), School 2 (424), School 3 (326), School 4 (252). Program mix is Tutoring (771), Counseling (382), and Enrichment (370).

In [7]:
```
# join school & track onto attendance
att_enriched = att.merge(
```

```
    students[["student_id","school_id","program_track"]],
    on="student_id",
    how="left"
)

by_school = (att_enriched.groupby("school_id")["attendance_rate"]
             .mean()
             .sort_index())

by_track = (att_enriched.groupby("program_track")["attendance_rate"]
            .mean()
            .sort_values(ascending=False))

display(by_school)
display(by_track)
```

```
school_id
1    0.807583
2    0.810650
3    0.797503
4    0.801477
Name: attendance_rate, dtype: float64
program_track
Enrichment    0.810115
Counseling    0.807257
Tutoring      0.801799
Name: attendance_rate, dtype: float64
```

- Average attendance rates are consistently around ~80% across schools (School 1 0.808, School 2 0.811, School 3 0.798, School 4 0.801), with small differences by program track (Enrichment 0.810, Counseling 0.807, Tutoring 0.802).

In [8]:
```
tmp0 = enrollments.copy()
tmp0["days_to_exit"] = (tmp0["exit_date"] - tmp0["enroll_date"]).dt.days

exit_rate = (tmp0["exit_date"].notna().mean())
early_exit_14d = ((tmp0["exit_date"].notna()) & (tmp0["days_to_exit"] <= 14)).mean()

print(f"Exit rate overall: {exit_rate:.3f}")
print(f"Early exit (<=14 days): {early_exit_14d:.3f}")
```

```
display(tmp0["days_to_exit"].describe())
```

```
Exit rate overall: 0.383
Early exit (<=14 days): 0.078
count    584.000000
mean      47.046233
std       29.421983
min        7.000000
25%       21.000000
50%       42.000000
75%       70.000000
max      112.000000
Name: days_to_exit, dtype: float64
```

- Overall exit rate is 38.3%. Among exited students (n = 584), the median time-to-exit is 42 days (IQR: 21–70), with an early exit (≤14 days) rate of 7.8% across all enrollments.
- Given a non-trivial exit rate (38.3%) and measurable early exits (7.8% within 14 days), cohort retention and early-risk flagging provide actionable levers for targeted outreach and program improvement.

From EDA to dashboard-ready tables

Based on the EDA findings (stable ~80% attendance with variation by school/track, and a non-trivial exit rate of 38.3% with early exits) and basic QA checks, I operationalized a small set of reporting metrics to support ongoing monitoring. I then created three dashboard-ready tables:

kpi_weekly.csv: weekly engagement and service volume (scheduled vs attended; weighted attendance rate)

student_risk.csv: student-level early warning flags (High/Medium/Low) with an explainable risk reason

cohort_retention.csv: cohort-based retention at 4 and 8 weeks by cohort month and school

These tables serve as the data sources for the Tableau dashboard sheets described below.

In [17]:
```
tmp = students.merge(
    enrollments[["student_id", "enroll_date", "exit_date", "status"]],
```

```
    on="student_id",
    how="left"
)

tmp["retained_4w"] = tmp["exit_date"].isna() | (
    tmp["exit_date"] >= tmp["enroll_date"] + pd.Timedelta(days=28)
)

tmp["retained_8w"] = tmp["exit_date"].isna() | (
    tmp["exit_date"] >= tmp["enroll_date"] + pd.Timedelta(days=56)
)

cohort_retention = (
    tmp.groupby(["cohort_month", "school_id"])
      .agg(
          cohort_size=("student_id", "count"),
          retention_4w=("retained_4w", "mean"),
          retention_8w=("retained_8w", "mean"),
      )
      .reset_index()
)

cohort_retention.to_csv(OUT_DIR / "cohort_retention.csv", index=False)
display(cohort_retention.head())
```

	cohort_month	school_id	cohort_size	retention_4w	retention_8w
0	2025-08-01	1	110	0.872727	0.763636
1	2025-08-01	2	90	0.900000	0.811111
2	2025-08-01	3	85	0.882353	0.800000
3	2025-08-01	4	44	0.931818	0.840909
4	2025-09-01	1	117	0.837607	0.683761

Risk flag rules:

- High: (scheduled_2w ≥ 2 AND attend_rate_2w < 0.60) OR (attend_rate_4w < 0.70)

- Medium: attend_rate_4w < 0.80
- Low: otherwise

In [23]:
```
today = att["week_start_date"].max()

last2 = att[att["week_start_date"] >= (today - pd.Timedelta(days=14))]
last4 = att[att["week_start_date"] >= (today - pd.Timedelta(days=28))]

l2 = (
    last2.groupby("student_id")
         .agg(attend_rate_2w=("attendance_rate", "mean"),
              scheduled_2w=("sessions_scheduled", "sum"))
         .reset_index()
)

l4 = (
    last4.groupby("student_id")
         .agg(attend_rate_4w=("attendance_rate", "mean"))
         .reset_index()
)

risk = (
    students[["student_id", "school_id", "cohort_month", "grade_level", "program_track"]]
    .merge(l2, on="student_id", how="left")
    .merge(l4, on="student_id", how="left")
)

def assign_risk(r: pd.Series) -> tuple[str, str]:
    if (
    pd.notna(r["scheduled_2w"])
    and r["scheduled_2w"] >= 2
    and pd.notna(r["attend_rate_2w"])
    and r["attend_rate_2w"] < 0.60
    ):
        return "High", "Low attendance (last 2 weeks)"
    if pd.notna(r["attend_rate_4w"]) and r["attend_rate_4w"] < 0.70:
        return "High", "Low attendance (last 4 weeks)"
    if pd.notna(r["attend_rate_4w"]) and r["attend_rate_4w"] < 0.80:
        return "Medium", "Moderate attendance risk"
    return "Low", "On track"
```

```
risk[["risk_level", "risk_reason"]] = risk.apply(lambda row: pd.Series(assign_risk(row)), axis=1)

risk.to_csv(OUT_DIR / "student_risk.csv", index=False)
display(risk["risk_level"].value_counts())
```

```
risk_level
Low       1424
High        80
Medium      19
Name: count, dtype: int64
```

In [19]:

```
kpi_weekly = (
    att.groupby("week_start_date")
       .agg(total_scheduled=("sessions_scheduled", "sum"),
            total_attended=("sessions_attended", "sum"))
       .reset_index()
)

kpi_weekly["attendance_rate"] = kpi_weekly["total_attended"] / kpi_weekly["total_scheduled"]

kpi_weekly.to_csv(OUT_DIR / "kpi_weekly.csv", index=False)
display(kpi_weekly.head())
```

	week_start_date	total_scheduled	total_attended	attendance_rate
0	2025-07-28	48	41	0.854167
1	2025-08-04	182	149	0.818681
2	2025-08-11	288	248	0.861111
3	2025-08-18	426	354	0.830986
4	2025-08-25	535	466	0.871028

Wrap-up

At this point, the raw operational tables have been transformed into three reporting-ready datasets (`kpi_weekly` , `cohort_retention` , and `student_risk`). These tables standardize KPI definitions and provide clean inputs for BI tools to support

ongoing monitoring and decision-making. In the next section, I use these outputs to build dashboard sheets for trend monitoring (attendance and service volume), early warning workflows (risk flags), and cohort-based performance tracking (4-week/8-week retention). The dashboard deliverables can be found in JeffreyLien_WorkSample2.